# DALAS Project
# Drug Repurposing

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## Contents

# 1 Introduction

## 1.1 Problem Statement

Clearly define the problem your project addresses. Explain its relevance in the scientific or practical context.

The discovery of new drugs is, although crucial, an extremely expensive and time-consuming process. Many candidate drugs ultimately turn out to be not only ineffective, but also potentially toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts' knowledge, it's important to be able to predict the likelihood of success for such studies. This project aims to explore the data and develop a potential prediction pipeline for this purpose. We acknowledge that industrial and academic drug development employ far more comprehensive methods, this project, by contrast, focuses on a preliminary, data-driven approach and is exploratory in nature.

## 1.2 Motivation and Objectives

Explain why this problem is important. State your main objectives and hypotheses.

The ability to predict the success of drug repurposing has academic and clinical value, it helps reducing the experimental cost and accelerating the discovery of effective treatments. The main objectives of this project are to:

- Explore public datasets to identify some relevant features
- Develop a preliminary machine learning pipeline for success prediction
- Evaluate the model performance
- Identify biases and potential ways for improvement

We decided to specify a group of diverse but closely related diseases - immune system disorders and drugs that are approved for at least one member of this group.

# 2 Data

## 2.1 Data Sources

Describe where the data comes from, collection methods, and aggregation process.

Among the vast amount of medical, pharmacological and chemical publicly available data, we focused our attention on those being able to provide us

Table 1: Description of data sources.

| **Source** | **Access Method** | **Data Retrieved** |
|---|---|---|
| MeSH RDF | SPARQL endpoint | Disease IDs and ontology |
| ChEMBL | REST API (Python client with query-style filters) | Drug chemical properties, targets and indications |
| UniProt | REST API (Python client) | Map target and pathway IDs |
| Open Targets | GraphQL API | Disease targets |
| ClinicalTrials.gov | REST API | Trial results and metadata |

## 2.2 Data Quality and Preprocessing

Discuss missing values, outliers, normalization, feature encoding, or transformations.

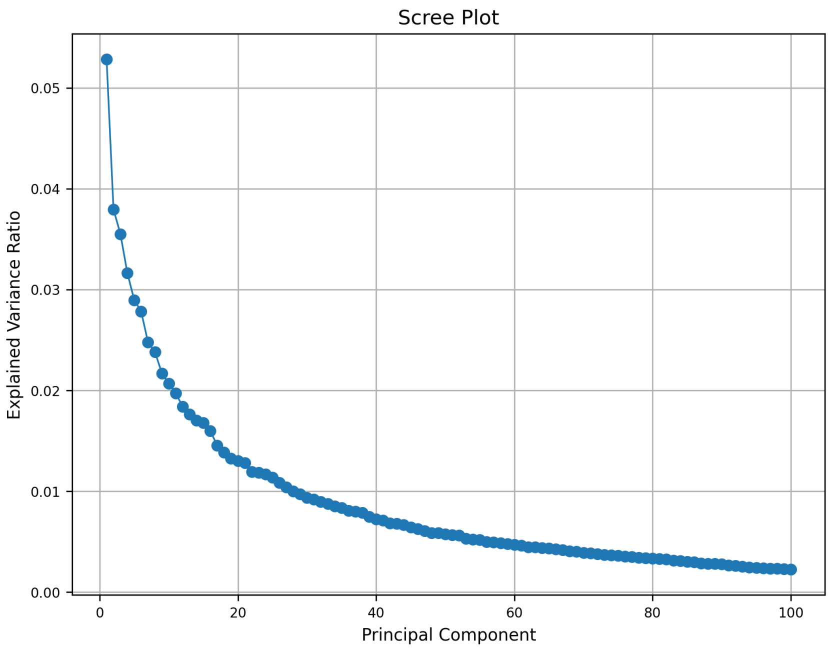


Figure 1: Scree plot of PCA on merged data.

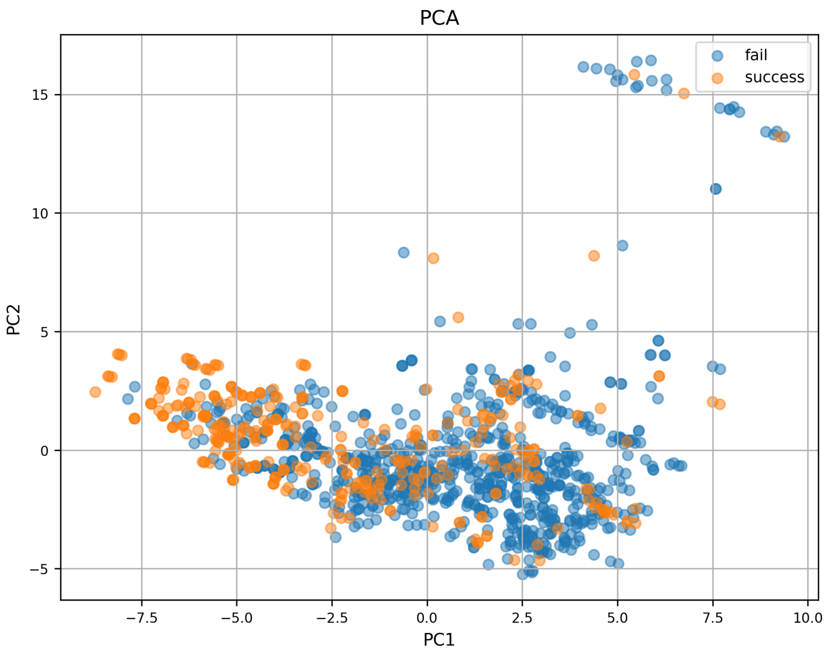


Figure 2: Scree plot of PCA on merged data.

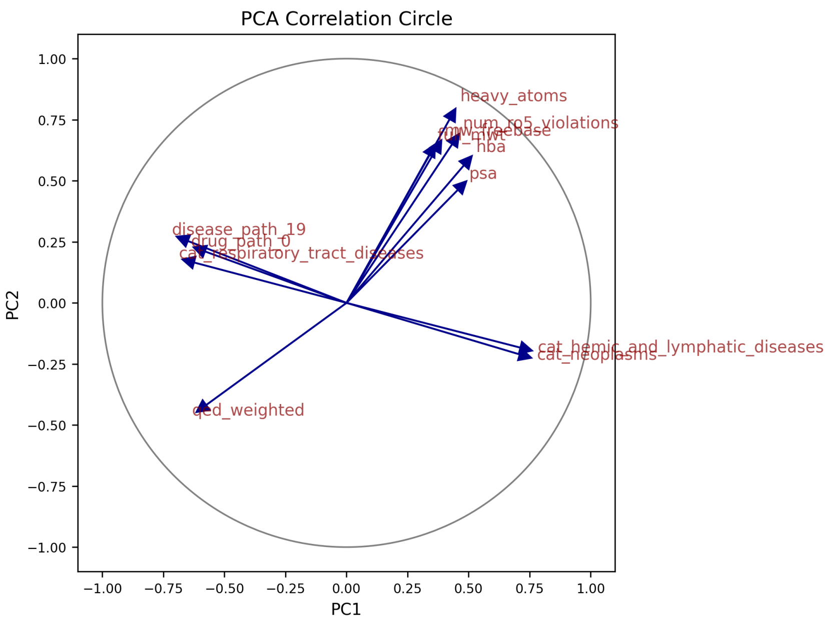


Figure 3: Scree plot of PCA on merged data.

## 2.3 Feature Description

Provide a table or list explaining each feature and its meaning. Optionally, include relationships between features.

Table 2: Summary of datasets and merged data.

| Dataset | Rows | Columns |
|---|---|---|
| Drugs | 541 | 181 |
| Diseases | 167 | 79 |
| Trials | 14335 | 12 |
| Indications | 2454 | 12 |
| Name embeddings | 489939 | 5 |
| Molecular fingerprints | 1029 | 101 |
| Merged data | 1142 | 253 |
| Merged data success counts | False: 801, True: 341 | |

Table 3: Description of features.

| Feature | Type | Description |
|---|---|---|
| **Drug** | | |
| drug_id | str | Unique ChEMBL identifier |
| chemical_probe | bool | Well-characterized and selective, typically used in biological research |
| dosed_ingredient | bool | Active ingredient that has been dosed in humans |
| inorganic_flag | bool | Inorganic molecule |
| natural_product | bool | Derived from a natural source (not fully synthetic) |
| oral | bool | Known to be administered orally |
| orphan | bool | Intended for use against a rare condition |
| parenteral | bool | Known to be administered parenterally |
| prodrug | bool | Prodrug, activated after administration |
| therapeutic_flag | bool | Has a therapeutic application, opposed to e.g. imaging |
| topical | bool | Known to be administered topically |
| veterinary | bool | Has veterinary application as well |
| chirality | cat | achiral,single enantiomer, mixture,nan |
| alogp | float | Octanol-water partition coefficient, lipophilicity indicator |
| aromatic_rings | int | Number of aromatic rings |
| full_mwt | float | Molecular weight of the full compound including any salts |
| hba | float | Number of hydrogen bond acceptors |
| hbd | float | Number of hydrogen bond donors |
| heavy_atoms | int | Number of heavy (non-hydrogen) atoms |
| mw_freebase | float | Molecular weight of parent compound |
| np_likeness_score | float | Natural product likeness score [Ertl et al., 2008] |
| num_ro5_violations | int | Violations of Lipinski's rule-of-five, using HBA and HBD definitions |
| psa | float | Polar surface area |
| qed_weighted | float | Weighted quantitative estimate of drug likeness [Bickerton et al., 2012] |
| ro3_pass | int | Passes the rule-of-three (mw < 300, logP < 3 etc) |
| rtb | int | Number of rotatable bonds |
| FP_# | float | First PCs in Morgan molecular fingerprint space |
| drug_path_# | float | First PCs in TF-IDF space derived from pathway lists |
| **Disease** | | |
| disease_id | str | Unique EFO identifier |
| ontology_depth | int | Depth in disease classification tree |
| nb_descendants | int | Number of descendants in disease classifications tree |
| nb_indications | int | Number of drug indications |
| cat_# | bool | Ontological category other than immune system disorders |
| disease_path_# | float | First PCs in TF-IDF space derived from pathway lists |
| **Drug–Disease** | | |
| success | bool | Consensus success or failure |
| path_similarity | float | Jaccard similarity of pathway lists |
| name_similarity | float | Cosine similarity of name embeddings in PubMedBERT (later excluded due to data leak) |

```
  HeteroData(
drug={ x=[742, 127] },
disease={ x=[167, 25] },
pathway={ x=[1509, 1] },
(drug, interacts, pathway)={ edge_index=[2, 31266] },
```

```
  (disease, associated_with, pathway)={ edge_index=[2, 13608] },
  (drug, tryed_against, disease)={
    edge_index=[2, 1298],
    edge_label=[1298],
  }
)
```

# 3 Methods

## 3.1 Model Overview

Briefly describe the models used. Include general principles (e.g., linear models, random forests, neural networks).

## 3.2 Implementation Details

Mention important parameters, software/libraries used, and cross-validation setup.

# 4 Results

## 4.1 Model Comparison

Compare models quantitatively (accuracy, RMSE, AUC, etc.) and qualitatively (strengths/weaknesses).

Table 4: Model comparison summary (random train test split).

| model | accuracy | precision | recall | f1_score | roc_auc | params | mean_test_score | std_test_score |
|---|---|---|---|---|---|---|---|---|
| XGBoost | 0.873362 | 0.782609 | 0.794118 | 0.788321 | 0.930307 | colsample_bytree = 0.8, gamma = 0.1, learning_rate = 0.05, max_depth = 5, n_estimators = 300, subsample = 0.8 | 0.920127 | 0.028953 |
| Support Vector Machine | 0.864629 | 0.746667 | 0.823529 | 0.783217 | NaN | C = 10, gamma = auto, kernel = rbf | 0.915037 | 0.022332 |
| Gradient Boost | 0.868996 | 0.796875 | 0.750000 | 0.772727 | 0.924096 | learning_rate = 0.05, max_depth = 5, n_estimators = 300, subsample = 0.8 | 0.918592 | 0.026515 |
| Random Forest | 0.851528 | 0.774194 | 0.705882 | 0.738462 | 0.928845 | max_depth = 20, min_samples_leaf = 1, min_samples_split = 2, n_estimators = 300 | 0.919067 | 0.028756 |
| k-Nearest Neighbors | 0.842795 | 0.742424 | 0.720588 | 0.731343 | 0.915373 | n_neighbors = 10, p = 1, weights = distance | 0.895602 | 0.027378 |
| Logistic Regression | 0.781659 | 0.597826 | 0.808824 | 0.687500 | 0.881714 | C = 0.1, l1_ratio = None, penalty = l2 | 0.858434 | 0.025389 |

Table 5: Model comparison summary (time based train test split).

| model | accuracy | precision | recall | f1_score | roc_auc | params | mean_test_score | std_test_score |
|---|---|---|---|---|---|---|---|---|
| Support Vector Machine | 0.799127 | 0.652778 | 0.691176 | 0.671429 | NaN | C = 10, gamma = auto, kernel = rbf | 0.794891 | 0.075004 |
| Random Forest | 0.799127 | 0.696429 | 0.573529 | 0.629032 | 0.836226 | max_depth = 10, min_samples_leaf = 2, min_samples_split = 2, n_estimators = 500 | 0.807810 | 0.061684 |
| Gradient Boost | 0.781659 | 0.645161 | 0.588235 | 0.615385 | 0.831202 | learning_rate = 0.1, max_depth = 3, n_estimators = 300, subsample = 0.8 | 0.814847 | 0.055349 |
| XGBoost | 0.790393 | 0.678571 | 0.558824 | 0.612903 | 0.805079 | colsample_bytree = 0.8, gamma = 0, learning_rate = 0.1, max_depth = 3, n_estimators = 300, subsample = 1.0 | 0.825033 | 0.069592 |
| Logistic Regression | 0.689956 | 0.484848 | 0.705882 | 0.574850 | 0.758129 | C = 0.01, l1_ratio = None, penalty = l2 | 0.789835 | 0.076975 |
| k-Nearest Neighbors | 0.681223 | 0.457627 | 0.397059 | 0.425197 | 0.714742 | n_neighbors = 20, p = 2, weights = distance | 0.789915 | 0.092087 |

## 4.2 Feature Analysis

Analyze the impact of key features on model performance (e.g., feature importance, correlation).

## 4.3 Failure and Success Cases

Highlight scenarios where the models perform well or poorly. Include illustrative figures or tables.

# 5 Discussion

Interpret the results. Relate findings to your original objectives. Discuss limitations and possible improvements.

# 6 Conclusion

Summarize your work, key findings, and potential future directions.

# References

[Bickerton et al., 2012] Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S., and Hopkins, A. L. (2012). Quantifying the chemical beauty of drugs. *Nat Chem*, 4(2):90–98.

[Ertl et al., 2008] Ertl, P., Roggo, S., and Schuffenhauer, A. (2008). Natural product-likeness score and its application for prioritization of compound libraries. *J. Chem. Inf. Model.*, 48(1):68–74. Epub 2007 Nov 23.